

Qining Zhang

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Ph.D. Candidate
Electrical Engineering and Computer Science
University of Michigan – Ann Arbor

Email: qiningz@umich.edu
Website: thumichzqn.github.io
[\[Google Scholar\]](#)

Research Interests

My research develops foundations and algorithms for reinforcement learning with human feedback, resource constraints, and representation bottlenecks, using stochastic zeroth-order policy optimization as a unified approach. My broader interests include reinforcement learning, bandits, adaptive experimentation, and stochastic optimization, with applications to language model post-training, queueing networks, robotics, and autonomous systems.

Education

Ph.D. Candidate, Electrical and Computer Engineering Aug. 2021 – May 2027 (expected)
M.S. in Mathematics May 2026

University of Michigan, Ann Arbor, MI, USA

Advisor: Prof. Lei Ying

Dissertation: *Stochastic Zeroth-Order Policy Gradient for Reinforcement Learning: Foundations, Algorithms, and Applications*

B.E., Electrical Information Science and Technology Aug. 2017 – Jun. 2021

Tsinghua University, Beijing, China

Advisors: Prof. Jintao Wang and Dr. Haoyue Tang

Thesis: *Online Algorithms for Optimizing Age of Synchronization in Complex Stochastic Networks*

Industry Experience

Applied Scientist Intern Jun. 2024 – Sept. 2024

Amazon.com (Search Experience Science), Seattle, WA, USA

Developed a multi-metric bandit algorithm to optimize Amazon.com Search pages through online adaptive experimentation, and deployed it to Amazon's internal A/B testing platform.

Honors and Awards

Rackham Predoctoral Fellowship, University of Michigan 2026

Notable Reviewer, ICLR 2025

Ph.D. Student Fellowship, University of Michigan 2021

Excellent Graduate of Tsinghua University 2021

Publications

Journals and Journal-Quality Conferences

- [1] **Qining Zhang**, Lei Ying. “Sign-SZPO: Provable Preference-Based Reinforcement Learning with an Unknown Link Function.” In *Reinforcement Learning Conference (RLC) and Reinforcement Learning Journal (RLJ)*, 2026. **(Acceptance Rate: 33.9%)** [\[paper\]](#)
- [2] **Qining Zhang**, Tanner Fiez, Yi Liu, Wenyang Liu. “Multi-Metric Adaptive Experimental Design Under a Fixed Budget with Validation.” In *Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026. **(Acceptance Rate: 28.0%)** [\[paper\]](#)
- [3] Yanxin Lin, Helene Hopfer, **Qining Zhang**, Misha T. Kwasniewski. “Fingerprinting and Quantification of Procyanidins via LC-MS/MS and ESI In-Source Fragmentation.” In *Journal of Agricultural and Food Chemistry*, 2025. [\[paper\]](#)
- [4] **Qining Zhang**, Lei Ying. “Zeroth-Order Policy Gradient for Reinforcement Learning from Human Feedback without Reward Inference.” In *International Conference on Learning Representations (ICLR)*, 2025. **(Acceptance Rate: 32.1%)** [\[paper\]](#)
- [5] Kellen Kanarios, **Qining Zhang**, Lei Ying. “Cost-Aware Best Arm Identification.” In *Reinforcement Learning Conference (RLC) and Reinforcement Learning Journal (RLJ)*, 2024. [\[paper\]](#)
- [6] **Qining Zhang**, Honghao Wei, Lei Ying. “Reinforcement Learning from Human Feedback without Reward Inference: Model-Free Algorithm and Instance-Dependent Analysis.” In *Reinforcement Learning Conference (RLC) and Reinforcement Learning Journal (RLJ)*, 2024. [\[paper\]](#)
- [7] Yifan Wang, **Qining Zhang**, Chuan Zhou, Lei Ying. “Deep Reinforcement Learning for Early Diagnosis of Lung Cancer.” In *AAAI Conference on Artificial Intelligence (AAAI)*, 2024. **(Acceptance Rate: 24.2%)** [\[paper\]](#)
- [8] **Qining Zhang**, Lei Ying. “Fast and Regret Optimal Best Arm Identification: Fundamental Limits and Low-Complexity Algorithms.” In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. **(Acceptance Rate: 26.1%)** [\[paper\]](#)
- [9] Yuchao Chen, Jintao Wang, **Qining Zhang**, Feifei Gao. “Online Utility Optimization in Multi-User Interference Networks Under a Long-Term Budget Constraint.” In *IEEE Transactions on Vehicular Technology*, 2022. [\[paper\]](#)
- [10] **Qining Zhang**, Honghao Wei, Weina Wang, Lei Ying. “On Low-Complexity Quickest Intervention of Mutated Diffusion Processes Through Local Approximation.” In *International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc)*, 2022. **(Acceptance Rate: 20.0%)** [\[paper\]](#)

Manuscripts Under Review

- [1] Deyi Wang*, **Qining Zhang***, Lei Ying. “Efficient Federated Reinforcement Learning from Human Feedback via Zeroth-Order Policy Optimization.” Under review, 2026. [\[paper\]](#)
- [2] Evan Assmus, **Qining Zhang**, Lei Ying. “SP3O: Reinforcement Learning from Segment Preferences without Reward Modeling.” Under review, 2026. [\[paper\]](#)
- [3] Josef Brozovich, **Qining Zhang**, Lei Ying. “Actor-Predictor Architecture for Reinforcement Learning with Formal Specification Objectives.” Under review, 2026.

* *Equal contribution.*

Other Publications

- [1] Yutong Wu, Yifan Wang, **Qining Zhang**, Chuan Zhou, Lei Ying. “Early Lung Cancer Diagnosis from Virtual Follow-Up LDCT Generation via Correlational Autoencoder and Latent Flow Matching.” In *W3PHIAI-26 Workshop at AAAI and Studies in Computational Intelligence*, 2025. [\[paper\]](#)
- [2] Yuchao Chen, Jintao Wang, **Qining Zhang**, Feifei Gao. “Online Optimizing Multi-User Interference Network Utility with Unknown CSI Under a Budget Constraint.” In *IEEE Wireless Communications and Networking Conference (WCNC)*, 2022. [\[paper\]](#)
- [3] **Qining Zhang**, Haoyue Tang, Jintao Wang. “Minimizing the Age of Synchronization in Power-Constrained Wireless Networks with Unreliable Time-Varying Channels.” In *IEEE Conference on Computer Communications Workshops (INFOCOM Workshops)*, 2020. [\[paper\]](#)

Patents

- [P1] Direct, Preference-Based Reinforcement Learning. Lei Ying, **Qining Zhang**. U.S. patent application.

Selected Grant and Proposal Writing Experience

- **Toyota Research Institute, HCAI-University Research Project**, 2026. *A Feedback-to-Foresight Framework to Facilitate Human–AI Co-Evolution through Reinforcement Learning: Theories and Applications*. Primary drafter: prepared the initial complete research proposal for subsequent revision.
- **Qualcomm Innovation Fellowship**, 2026. *Federated Reinforcement Learning from Human Feedback in Networked Edge Systems*. Primary drafter: building on our prior work on federated RLHF; currently extending the project into an NSF grant proposal.
- **Rackham Predoctoral Fellowship, University of Michigan**, 2026. Prepared the successful research proposal for the fellowship application.

Research Mentoring

Current Students

- Yifan Jing, undergraduate (Summer 2026 – Present), research topic: Enhancing LM Interaction through Automatic Prompt Injection.
- Evan Assmus, undergraduate (Fall 2024 – Present), research topic: Segment-Based Policy Optimization for Reinforcement Learning.

Former Students

- Yutong Wu, undergraduate (Winter 2025 – Fall 2026), research topic: Latent Diffusion Models for Lung Cancer Early Diagnosis; currently a graduate student at Stanford University.
- Chenshun Ni, master's student (Winter 2024 – Fall 2024), research topic: Diffusion Models for Medical Imaging; currently a Ph.D. student at Ohio State University.
- Shihui Sun, undergraduate (Winter 2024 – Fall 2024), research topic: Misinformation Intervention in Social Networks.
- Wentao Wei, undergraduate (Winter 2024 – Fall 2024), research topic: Misinformation Intervention in Social Networks.
- Kellen Kanarios, undergraduate (Fall 2022 – Winter 2024), research topic: Cost-Aware Best Arm Identification; currently a Ph.D. student at the University of Michigan.
- Jinheng Zhang, undergraduate (Summer 2022), research topic: Bayesian Offline Multi-Armed Bandits; currently a Ph.D. student at the University of Pennsylvania.

Teaching Experience

Graduate Student Instructor

Fall 2023 – Summer 2026

SIADS 644: Reinforcement Learning Algorithms

University of Michigan, Ann Arbor, MI, USA

- Held office hours and recitations on core reinforcement learning concepts; provided support including debugging and assignment/conceptual questions.
- Developed new instructional materials on reinforcement learning from human feedback.

Invited Talks

- [1] Provable Preference-Based Reinforcement Learning with an Unknown Link Function
– Invited Oral Talk at *SIGMETRICS Workshop*, Jun. 2026.

- [2] Zeroth-Order Policy Gradient for Reinforcement Learning from Human Feedback without Reward Inference
 - Invited Oral Talk at *INFORMS Annual Meeting*, Oct. 2025.
- [3] Fast and Regret Optimal Best Arm Identification: Fundamental Limits and Low-Complexity Algorithms
 - Invited 60-Minute Oral Talk at *Amazon Prime Science Seminar*, Feb. 2024.
 - Invited Oral Talk at *Annual Allerton Conference*, Sept. 2023.

Conference and Workshop Presentations

- [1] Provable Preference-Based Reinforcement Learning with an Unknown Link Function
 - Oral Talk and Poster at *Reinforcement Learning Conference (RLC)*, Aug. 2026.
 - Poster at *RLHF Workshop of IMSI*, Apr. 2026.
 - Poster at *ML×OR Workshop of NeurIPS*, Dec. 2025.
- [2] Zeroth-Order Policy Gradient for Reinforcement Learning from Human Feedback without Reward Inference
 - Poster at *International Conference on Learning Representations (ICLR)*, Apr. 2025.
- [3] Reinforcement Learning from Human Feedback without Reward Inference: Model-Free Algorithm and Instance-Dependent Analysis
 - Oral Talk and Poster at *Reinforcement Learning Conference (RLC)*, Aug. 2024.
- [4] Fast and Regret Optimal Best Arm Identification: Fundamental Limits and Low-Complexity Algorithms
 - Poster at *Conference on Neural Information Processing Systems (NeurIPS)*, Dec. 2023.
- [5] On Low-Complexity Quickest Intervention of Mutated Diffusion Processes: From Local Approximation to Misinformation and Anomaly Detection
 - Oral Talk at *International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc)*, Oct. 2022.
- [6] Minimizing the Age of Synchronization in Power-Constrained Wireless Networks with Unreliable Time-Varying Channels
 - Oral Talk at *Age-of-Information Workshop of IEEE Conference on Computer Communications (INFOCOM)*, Jul. 2020.

Professional Service

- Journal Reviewer: IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI; 2024 Reviewer Certificate), IEEE/ACM Transactions on Networking, Automatica, Performance Evaluation, Transactions on Machine Learning Research (TMLR).
- Conference Reviewer: ICLR (2025, 2026), INFOCOM 2025, L4DC 2025, RLC (2025, 2026), NeurIPS 2025, AISTATS 2026, EWRL 2026, AAAI 2027.
- Co-organizer, Michigan RL Seminar, University of Michigan, 2024; a semester-long reinforcement learning seminar with four invited speakers.
- Co-organizer, DTV Lab Seminar Series, Tsinghua University, 2021.